

ECE 332: Engineering Electromagnetics II

Lecture 18

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June 2, 2026

Linear Antenna Arrays (Continued)

Since Q is in the far field of all of the antennas, it's possible to make 2 approximations in the expression for $\tilde{E}(R_0, \theta, \phi)$.

- 1 For R_i in the denominators of the spherical propagation factors,

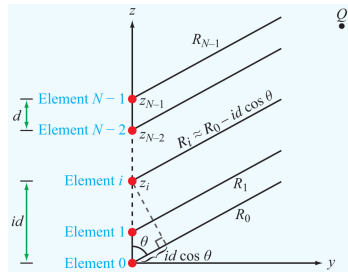
$$R_i \approx R_0 \quad (1)$$

- 2 Assume that the line segments connecting Antenna i to Q are parallel for all i . This leads to the approximation

$$e^{-jkR_i} \approx e^{-jk(R_0 - z_i)} = e^{-jk(R_0 - id \cos \theta)} = e^{-jkR_0} e^{jkid \cos \theta}. \quad (2)$$

Applying these approximations,

$$\tilde{E}(R_0, \theta, \phi) = \left(\sum_{i=0}^{N-1} A_i \frac{e^{-jkR_i}}{R_i} \right) \tilde{f}_e(\theta, \phi) = \tilde{f}_e(\theta, \phi) \left(\frac{e^{-jkR_0}}{R_0} \right) \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta}. \quad (3)$$



Linear Antenna Arrays (Continued)

Corresponding to the phasor of the electric field

$$\tilde{E}(R_0, \theta, \phi) = \tilde{f}_e(\theta, \phi) \left(\frac{e^{-jkR_0}}{R_0} \right) \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta}, \quad (4)$$

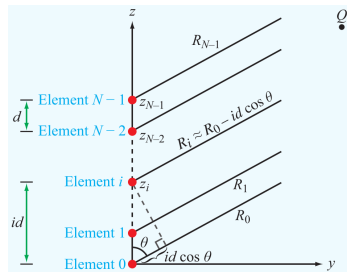
the average power density of the array is

$$\begin{aligned} S(R_0, \theta, \phi) &= \frac{1}{2\eta_0} |\tilde{E}(R_0, \theta, \phi)|^2 \\ &= \frac{|\tilde{f}_e(\theta, \phi)|^2}{2\eta_0 R_0^2} \left| \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right|^2 \\ &= S_e(R_0, \theta, \phi) F_a(\theta). \end{aligned} \quad (5)$$

Here,

$$S_e(R_0, \theta, \phi) = \frac{|\tilde{f}_e(\theta, \phi)|^2}{2\eta_0 R_0^2} \quad (6)$$

is the **average power density due to an individual element**, as before,



Linear Antenna Arrays (Continued)

and

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right|^2 \quad (7)$$

is the **array factor**, which is the far-field radiation intensity of the array with isotropic radiators in place of the antennas.

The equation

$$S(R_0, \theta, \phi) = S_e(R_0, \theta, \phi) F_a(\theta) \quad (8)$$

implies that the overall power density can be determined by multiplying

- the far-field power pattern $F_a(\theta)$ for a corresponding array of isotropic radiators
- and the power density $S_e(R_0, \theta, \phi)$ for an individual antenna of the type used in the array.

This is called the **pattern multiplication principle**.

Linear Antenna Arrays (Continued)

Using the polar forms for the feeding coefficients, the array factor can be written as

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} a_i e^{j\psi_i} e^{jkid \cos \theta} \right|^2. \quad (9)$$

Therefore, the array factor (and, hence, the antenna pattern) is determined by the **array amplitude distribution**

$$a_0, a_1, \dots, a_{N-1} \quad (10)$$

and the **array phase distribution**

$$\psi_0, \psi_1, \dots, \psi_{N-1}. \quad (11)$$

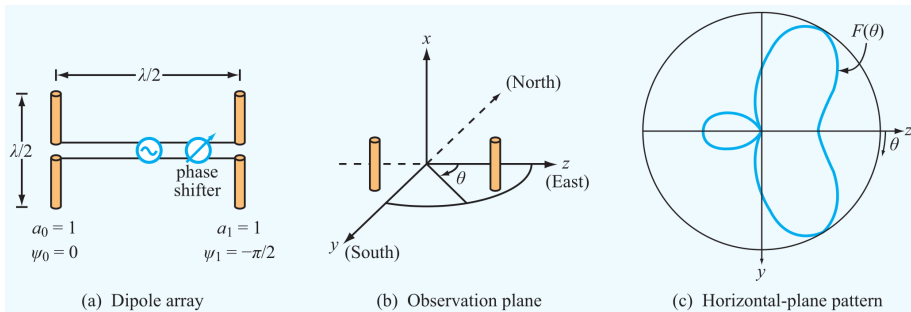
More specifically,

- the *array amplitude distribution* controls the **shape**
- and the *array phase distribution* controls the **direction**

of the array antenna pattern.

Array of Two Half-Wave Dipoles

Consider the array of two half-wave dipole antennas intended for transmitting EM waves with wavelength λ placed a distance $\lambda/2$ apart.



The z -axis passes through the centers of the antennas at a height $\lambda/4$ above the ground. (The ground is the plane $x = -\lambda/4$.) Both antennas are oriented along the x -direction. One antenna is located at $z = -\lambda/4$ and the other one is located at $z = \lambda/4$. (Different than assumed before.)

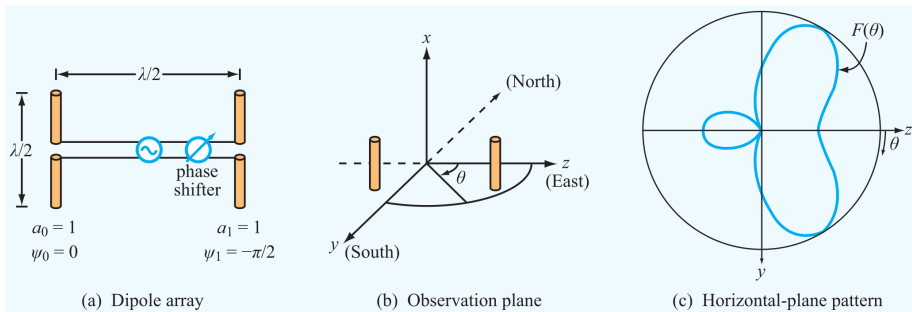
Array of Two Half-Wave Dipoles (Continued)

The **amplitudes of the feeding coefficients** are

$$a_0 = a_1 = 1 \quad (12)$$

and there is a phase shift of $-\pi/2$ radians between the emitted signals so the **phases of the feeding coefficients** are

$$\psi_0 = 0 \quad \text{and} \quad \psi_1 = -\pi/2. \quad (13)$$

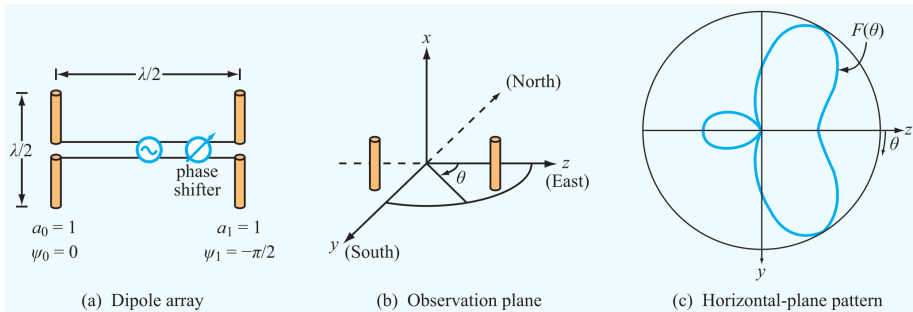


Array of Two Half-Wave Dipoles (Continued)

Setting $\psi_0 = 0$ and $\psi_1 = -\pi/2$ amounts to measuring the phase relative to the phase of Antenna 0. **Phases are usually measured**

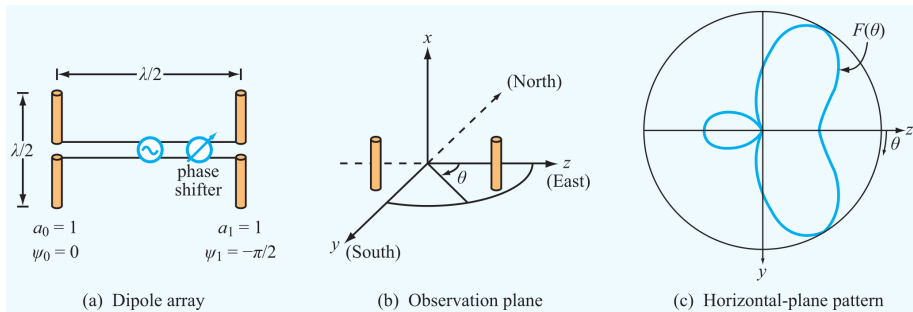
- relative to the phase of one antenna
- or relative to the phase of the source.

Notice that the dipoles are **oriented differently** than in the previous analysis. In the current configuration, the angle θ sweeps a **plane perpendicular to the axes of the dipoles**, like ϕ previously.



Array of Two Half-Wave Dipoles (Continued)

So, in a horizontal plane, parallel to the y - z plane, an individual half-wave dipole radiates uniformly at a value that only depends on the specific plane and R and doesn't depend on θ .



We'll consider the y - z plane, specifically. In this plane, the **average power density of an individual half-wave dipole** is

$$S_e = S_0 = \frac{15I_0^2}{\pi R^2} \quad (14)$$

Array of Two Half-Wave Dipoles (Continued)

and

$$S(R, \theta) = S_0 F_a(\theta). \quad (15)$$

In this case,

$$k = 2\pi/\lambda, \quad d = \lambda/2 \quad \implies \quad kd = \pi, \quad (16)$$

and the **array factor** is

$$\begin{aligned} F_a(\theta) &= \left| \sum_{i=0}^1 a_i e^{j\psi_i} e^{jkid \cos \theta} \right|^2 = \left| a_0 e^{j\psi_0} e^{j0} + a_1 e^{j\psi_1} e^{j\pi \cos \theta} \right|^2 \\ &= \left| 1 + e^{-j\pi/2} e^{j\pi \cos \theta} \right|^2 = \left| 1 + e^{j(\pi \cos \theta - \pi/2)} \right|^2. \end{aligned} \quad (17)$$

It's possible to express this in a more familiar form. For any value x ,

$$\begin{aligned} 1 + e^{jx} &= e^{jx/2} \left(e^{-jx/2} + e^{jx/2} \right) = 2e^{jx/2} \left(\frac{e^{jx/2} + e^{-jx/2}}{2} \right) \\ &= 2e^{jx/2} \cos(x/2). \end{aligned} \quad (18)$$

Array of Two Half-Wave Dipoles (Continued)

Let $x = \pi \cos \theta - \pi/2$. Then, the array factor is

$$\begin{aligned} F_a(\theta) &= \left| 1 + e^{jx} \right|^2 = \left| 2e^{jx/2} \cos(x/2) \right|^2 = 4 \left| e^{jx/2} \right|^2 \left| \cos(x/2) \right|^2 \\ &= 4 \cos^2(x/2) = 4 \cos^2 \left(\frac{\pi}{2} \cos \theta - \frac{\pi}{4} \right). \end{aligned} \quad (19)$$

Altogether, the **average power density of the array** in the y - z plane is

$$S(R, \theta) = 4S_0 \cos^2 \left(\frac{\pi}{2} \cos \theta - \frac{\pi}{4} \right). \quad (20)$$

The function $\cos^2 \alpha$ has maxima when $\cos \alpha = \pm 1$, i.e. when $\alpha = m\pi$ for any integer m . So, if there is a solution to the equation

$$m\pi = \frac{\pi}{2} \cos \theta - \frac{\pi}{4} \quad (21)$$

for some integer m , the power density reaches a maximum value of $4S_0$.

Array of Two Half-Wave Dipoles (Continued)

Rearranging this equation and multiplying it by $2/\pi$ implies

$$\cos \theta = 2m + \frac{1}{2}. \quad (22)$$

For $m \neq 0$, $|2m + 1/2| > 1$. So, the only possible solution occurs when $m = 0$ and

$$\cos \theta = \frac{1}{2} \implies \theta = \frac{\pi}{3} + 2\pi n, \frac{5\pi}{3} + 2\pi p \quad (23)$$

for any integers n and p . In the current situation, the angles satisfying $0 \leq \theta < 2\pi$ cover the entire plane. Within this range, the **array factor reaches a maximum value of 4** when

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}. \quad (24)$$

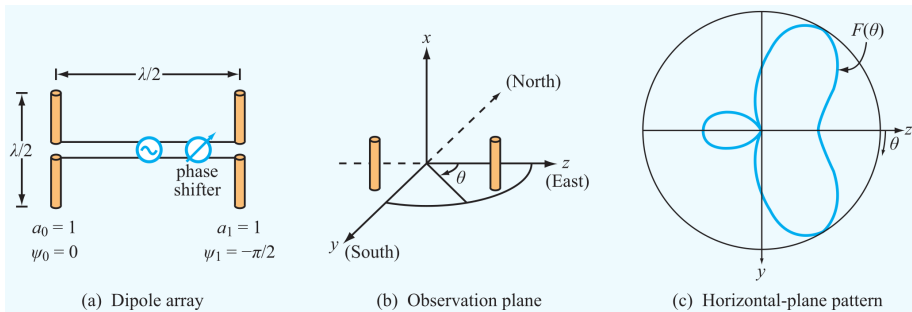
So, in the y - z plane, the **maximum power density of the array** is

$$S_{\max} = 4S_0 \quad (25)$$

Array of Two Half-Wave Dipoles (Continued)

and the **normalized radiation intensity of the array** is

$$F(\theta) = \frac{S(R, \theta)}{S_{\max}} = \cos^2 \left(\frac{\pi}{2} \cos \theta - \frac{\pi}{4} \right). \quad (26)$$



Module 9.5 on the book's website allows you to plot the antenna pattern for a two-dipole array with different values for the array parameters and Example 9-6 considers the inverse problem of finding the parameters of the array that result in a specific array antenna pattern.

N-Element Array with Equal Phase

Consider the **original N-element array** along the z-axis with **equal spacing** d between neighboring elements and **equal phases**, i.e.

$$\psi_i = \psi_0 \text{ for all } i. \quad (27)$$

This is called a **broadside array**. In this case, the **array factor** is

$$\begin{aligned} F_a(\theta) &= \left| \sum_{i=0}^{N-1} a_i e^{j\psi_i} e^{jkid \cos \theta} \right|^2 = \left| \sum_{i=0}^{N-1} a_i e^{j\psi_0} e^{jkid \cos \theta} \right|^2 \\ &= \left| e^{j\psi_0} \sum_{i=0}^{N-1} a_i e^{jkid \cos \theta} \right|^2 = |e^{j\psi_0}|^2 \left| \sum_{i=0}^{N-1} a_i e^{jkid \cos \theta} \right|^2 \\ &= \left| \sum_{i=0}^{N-1} a_i e^{jkid \cos \theta} \right|^2, \end{aligned} \quad (28)$$

N-Element Array with Equal Phase (Continued)

i.e.

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} a_i e^{jkid \cos \theta} \right|^2. \quad (29)$$

So, the **phase difference between neighboring elements** is entirely due to the distance between them and the direction θ and is

$$\gamma = kd \cos \theta = \frac{2\pi d}{\lambda} \cos \theta. \quad (30)$$

(This γ again has no relationship to the propagation constant.) With this definition, the **array factor** is

$$F_a(\gamma) = \left| \sum_{i=0}^{N-1} a_i e^{ji\gamma} \right|^2. \quad (31)$$

Now, suppose that **all of the amplitudes are the same**, as well, i.e.

$$a_i = a \in \mathbb{R} \text{ for all } i \text{ with } a > 0. \quad (32)$$

N-Element Array with Equal Phase (Continued)

With equal phases and equal amplitudes, the array factor is

$$F_a(\gamma) = a^2 \left| \sum_{i=0}^{N-1} e^{ji\gamma} \right|^2 = a^2 \left| 1 + e^{j\gamma} + e^{j2\gamma} + \dots + e^{j(N-1)\gamma} \right|^2. \quad (33)$$

Let

$$f_a(\gamma) = 1 + e^{j\gamma} + e^{j2\gamma} + \dots + e^{j(N-1)\gamma} \quad (34)$$

so that

$$F_a(\gamma) = a^2 |f_a(\gamma)|^2. \quad (35)$$

Here,

$$f_a(\gamma) = (e^{j\gamma})^0 + (e^{j\gamma})^1 + (e^{j\gamma})^2 + \dots + (e^{j\gamma})^{N-1}, \quad (36)$$

which is a finite geometric series of the form $\sum_{n=0}^{N-1} r^n$ with $r = e^{j\gamma}$.

As shown in the book (or any calculus textbook),

$$\sum_{n=0}^{N-1} r^n = \frac{1 - r^N}{1 - r} \quad (37)$$

N -Element Array with Equal Phase (Continued)

and

$$\begin{aligned} f_a(\gamma) &= \frac{1 - e^{jN\gamma}}{1 - e^{j\gamma}} = \frac{e^{jN\gamma/2}}{e^{j\gamma/2}} \left(\frac{e^{-jN\gamma/2} - e^{jN\gamma/2}}{e^{-j\gamma/2} - e^{j\gamma/2}} \right) \\ &= e^{j(N-1)\gamma/2} \frac{\sin(N\gamma/2)}{\sin(\gamma/2)}. \end{aligned} \quad (38)$$

Therefore,

$$\begin{aligned} F_a(\gamma) &= a^2 |f_a(\gamma)|^2 = a^2 \left| e^{j(N-1)\gamma/2} \frac{\sin(N\gamma/2)}{\sin(\gamma/2)} \right|^2 \\ &= a^2 \left| e^{j(N-1)\gamma/2} \right|^2 \left| \frac{\sin(N\gamma/2)}{\sin(\gamma/2)} \right|^2 \\ &= a^2 \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}, \end{aligned} \quad (39)$$

N-Element Array with Equal Phase (Continued)

i.e. **when the amplitudes and phases are equal** the array factor is

$$F_a(\gamma) = a^2 \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}. \quad (40)$$

Notice that

$$\begin{aligned} |f_a(\gamma)| &= |1 + e^{j\gamma} + e^{j2\gamma} + \dots + e^{j(N-1)\gamma}| \\ &\leq |1| + |e^{j\gamma}| + |e^{j2\gamma}| + \dots + |e^{j(N-1)\gamma}| \\ &= N. \end{aligned} \quad (41)$$

When

$$kd \cos \theta = \gamma = 0, \quad \text{i.e. } \theta = \frac{\pi}{2}, \quad (42)$$

$|f_a(\gamma)|$ has this maximum value and

$$F_a(\gamma) = F_a(0) = a^2 |f_a(0)|^2 = a^2 N^2 = F_{a,\max}. \quad (43)$$

N-Element Array with Equal Phase (Continued)

Therefore, the **normalized array factor** is

$$\begin{aligned} F_{an} &= \frac{F_a(\gamma)}{F_{a,\max}} = \frac{1}{a^2 N^2} \left(a^2 \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)} \right) \\ &= \frac{\sin^2(N\gamma/2)}{N^2 \sin^2(\gamma/2)} \\ &= \frac{\sin^2 \left(\frac{N\pi d}{\lambda} \cos \theta \right)}{N^2 \sin^2 \left(\frac{\pi d}{\lambda} \cos \theta \right)}. \end{aligned} \quad (44)$$

This could be the same as the antenna pattern but isn't always since the power density has an additional factor of $S_e(R_0, \theta, \phi)$, which is the power density of an individual element.

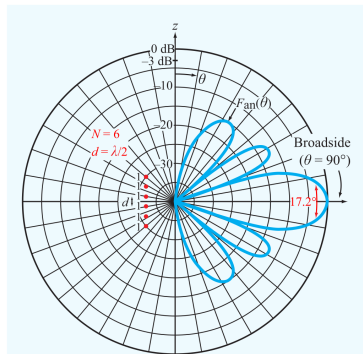


Figure 9-30 Normalized array pattern of a uniformly excited six-element array with interelement spacing $d = \lambda/2$.

Electronically Steering Antenna Arrays

As advertised earlier, an antenna array can be **electronically steered**. This means **using the phase differences between elements to change the direction θ of the main lobe of the array**. Note that some arrays can also be **mechanically steered** and some arrays support **both types of steering**. In fact, some arrays of radio telescopes, such as the Very Large Array (VLA, [Wikipedia](#)) in New Mexico and the Atacama Large Millimeter/submillimeter Array (ALMA, [Wikipedia](#)) in Chile, allow entire radio dishes to be relocated to change

- the distance between elements
- and the overall array geometry.

A linear phase distribution

$$\psi_i = -i\delta \quad (45)$$

for some constant **incremental phase delay** δ modifies the **array factor** $F_a(\theta)$ and allows an array to be electronically steered. An array with

Electronically Steering Antenna Arrays (Continued)

this type of phase distribution is called a **linearly phased array**.

If $\psi_i = -i\delta$, then the array factor is

$$\begin{aligned} F_a(\theta) &= \left| \sum_{i=0}^{N-1} a_i e^{j\psi_i} e^{jkid \cos \theta} \right|^2 = \left| \sum_{i=0}^{N-1} a_i e^{-ji\delta} e^{jkid \cos \theta} \right|^2 \\ &= \left| \sum_{i=0}^{N-1} a_i e^{ji(kd \cos \theta - \delta)} \right|^2 = \left| \sum_{i=0}^{N-1} a_i e^{ji\gamma'} \right|^2, \end{aligned} \quad (46)$$

where

$$\gamma' = kd \cos \theta - \delta. \quad (47)$$

It's useful to define

$$\delta = kd \cos \theta_0 = \frac{2\pi d}{\lambda} \cos \theta_0 \quad (48)$$

where θ_0 is called the **scan angle**. With this definition,

$$\gamma' = kd(\cos \theta - \cos \theta_0) = \frac{2\pi d}{\lambda}(\cos \theta - \cos \theta_0) \quad (49)$$

Electronically Steering Antenna Arrays (Continued)

and the array factor is

$$F_a(\theta) = F_a(\gamma') = \left| \sum_{i=0}^{N-1} a_i e^{ji\gamma'} \right|^2. \quad (50)$$

Recall that the array factor for the N -element array with equal phase was

$$F_a(\gamma) = \left| \sum_{i=0}^{N-1} a_i e^{ji\gamma} \right|^2. \quad (51)$$

So, the array factor for the array with a linear phase distribution **is the same as** the array factor for the array with equal phase (with the same amplitude distribution) with γ' replacing γ .

Notice that

$$\left| \sum_{i=0}^{N-1} a_i e^{ji\gamma'} \right| \leq \sum_{i=0}^{N-1} |a_i e^{ji\gamma'}| = \sum_{i=0}^{N-1} |a_i| |e^{ji\gamma'}| = \sum_{i=0}^{N-1} |a_i|. \quad (52)$$

Electronically Steering Antenna Arrays (Continued)

If $\gamma' = 0$, then

$$\left| \sum_{i=0}^{N-1} a_i e^{ji\gamma'} \right| = \left| \sum_{i=0}^{N-1} a_i \right| = \sum_{i=0}^{N-1} a_i = \sum_{i=0}^{N-1} |a_i|, \quad (53)$$

since $a_i \in \mathbb{R}$ with $a_i \geq 0$ for all i . So, **when** $\gamma' = 0$, i.e.

$$\cos \theta = \cos \theta_0 \implies \theta = \theta_0 \text{ for } 0 \leq \theta \leq \pi, \quad (54)$$

$F_a(\gamma')$ **reaches its maximum value**. (Typically, there will be other local maxima and some of them could have the global maximum value, too. See Figure 9-31 on p. 421, for example.) So, setting the **incremental phase delay in the linear phase distribution** to

$$\delta = \frac{2\pi d}{\lambda} \cos \theta_0 \quad (55)$$

steers the maximum power density to $\theta = \theta_0$.

Electronically Steering Antenna Arrays (Continued)

There are **two special directions** worth identifying. The **maximum power density of the array** occurs

- in the **broadside direction** $\theta_0 = \pi/2$ when $\delta = 0$,
- in the **end-fire direction** $\theta_0 = 0$ when $\delta = kd \cos 0 = kd = 2\pi d/\lambda$.

When the **amplitudes are equal**, the **normalized array factor** is

$$\begin{aligned} F_{an} &= \frac{\sin^2(N\gamma'/2)}{N^2 \sin^2(\gamma'/2)} \\ &= \frac{\sin^2\left(\frac{N\pi d}{\lambda}(\cos \theta - \cos \theta_0)\right)}{N^2 \sin^2\left(\frac{\pi d}{\lambda}(\cos \theta - \cos \theta_0)\right)}. \end{aligned} \quad (56)$$

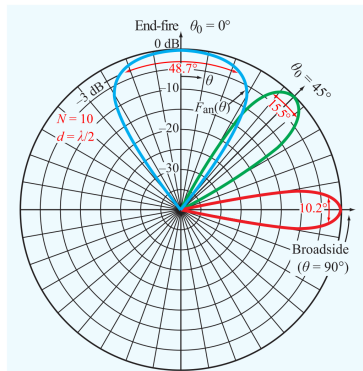


Figure 9-33 Normalized array pattern of a 10-element array with $\lambda/2$ spacing between adjacent elements. All elements are excited with equal amplitude. Through the application of linear phase across the array, the main beam can be steered from the broadside direction ($\theta_0 = 90^\circ$) to any scan angle θ_0 . Equiphasic excitation corresponds to $\theta_0 = 90^\circ$.

Array Feeding and Frequency Scanning

Steering the maximum power density of an antenna array to an angle θ_0 requires **creating a linear phase distribution**

$$\psi_i = -i\delta \quad (57)$$

across the array in which the **incremental phase delay** satisfies

$$\delta = kd \cos \theta_0 = \frac{2\pi d}{\lambda} \cos \theta_0. \quad (58)$$

One approach to meeting these requirements is based on using

- the **lengths of the transmission lines** that feed the elements of the array **to produce the phase delay**
- and **varying the frequency** of the signal sent to the antennas over a narrow frequency range

$$f_0 - \Delta f \leq f \leq f_0 + \Delta f, \quad (59)$$

where f_0 is the **reference frequency** intended to be transmitted.

Array Feeding and Frequency Scanning (Continued)

Controlling the direction of the main lobe by varying the frequency is called **frequency scanning**. This is why θ_0 was called the **scan angle**.

Suppose the **path length** for the transmission line connected to Element i is

$$\ell_i = \ell_0 + i\Delta\ell, \quad (60)$$

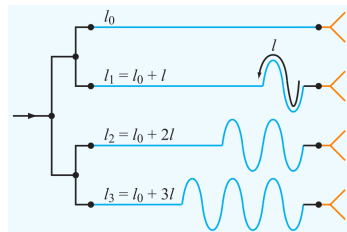
where

- ℓ_0 is the path length for Element 0
- and $\Delta\ell$ is a constant length increment.

The **phase factor** $e^{-j\beta\ell_i}$, where

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi f}{u_p} \quad (61)$$

is the **phase constant**, represents EM waves of frequency f propagating through a transmission line of length ℓ_i . Here u_p is the **phase velocity**.



Array Feeding and Frequency Scanning (Continued)

If the phase is measured **relative to the phase of Element 0**, which means ψ_0 is set to 0, the phase of Element i is

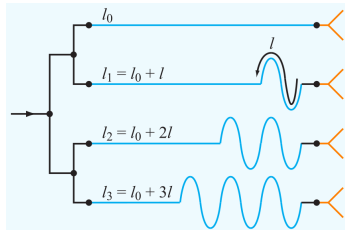
$$\begin{aligned}\psi_i(f) &= -\beta \ell_i - (-\beta \ell_0) = -\beta(\ell_i - \ell_0) = -\beta i \Delta \ell \\ &= -\frac{2\pi}{\lambda} i \Delta \ell = -\frac{2\pi i \Delta \ell}{u_p} f.\end{aligned}\quad (62)$$

Let the **length increment** be

$$\Delta \ell = n_0 \lambda_0 = \frac{n_0 u_p}{f_0}, \quad (63)$$

where n_0 is a **positive integer** to be chosen in the design. Using this value for $\Delta \ell$, the **phase of Element i at frequency f** is

$$\psi_i(f) = -\frac{2\pi i}{u_p} f \left(\frac{n_0 u_p}{f_0} \right) = -2in_0\pi \left(\frac{f}{f_0} \right). \quad (64)$$



Array Feeding and Frequency Scanning (Continued)

Reference Frequency

At the **reference frequency**, $f = f_0$ and the **phase of Element i** is

$$\psi_i(f_0) = -2in_0\pi, \quad (65)$$

which is an integer multiple of -2π . Therefore, at the reference frequency

$$e^{j\psi_i(f_0)} = e^{-j2in_0\pi} = 1 = e^0 = e^{j\psi_0} \text{ for all } i, \quad (66)$$

which means that **all elements have phase equivalent to** $\psi_0 = 0$ and

$$kd \cos \theta = 0 \implies \cos \theta = 0 \implies \theta = \frac{\pi}{2}. \quad (67)$$

So, at the reference frequency the array is a broadside array.

Array Feeding and Frequency Scanning (Continued)

Frequency $f = f_0 + \Delta f$

At the **frequency** $f = f_0 + \Delta f$, the **phase of Element** i is

$$\begin{aligned}\psi_i(f_0 + \Delta f) &= -2in_0\pi \left(\frac{f_0 + \Delta f}{f_0} \right) = -2in_0\pi - 2in_0\pi \left(\frac{\Delta f}{f_0} \right) \\ &= -2in_0\pi - i\delta,\end{aligned}\tag{68}$$

where the **incremental phase delay** is

$$\delta = 2n_0\pi \left(\frac{\Delta f}{f_0} \right).\tag{69}$$

Since

$$e^{j\psi_i} = e^{j(-2in_0\pi - i\delta)} = e^{-2in_0\pi} e^{-ji\delta} = e^{-ji\delta},\tag{70}$$

the phase $\psi_i(f_0 + \Delta f)$ is equivalent to $-i\delta$. The corresponding **scan angle** θ_0 satisfies

$$kd \cos \theta_0 = \delta = 2n_0\pi \left(\frac{\Delta f}{f_0} \right) \implies \cos \theta_0 = \frac{2n_0\pi}{kd} \left(\frac{\Delta f}{f_0} \right) = \frac{n_0\lambda}{d} \left(\frac{\Delta f}{f_0} \right).\tag{71}$$

Array Feeding and Frequency Scanning (Continued)

Although

$$k = \frac{2\pi f}{u_p} \quad \text{and} \quad \lambda = \frac{u_p}{f} \quad (72)$$

both depend on the frequency $f = f_0 + \Delta f$, it's possible to **assume**

$$k = k_0 = \frac{2\pi f_0}{u_p} \quad \text{and} \quad \lambda = \lambda_0 = \frac{u_p}{f_0} \quad (73)$$

and only introduce an **error on the order of $\Delta f/f_0$ in $\cos \theta_0$** , which is reasonable if Δf is small enough.

For a **desired scan angle θ_0** and **fixed values for d and f_0** , choosing a **smaller Δf** means having to choose a **larger value for n_0** and vice versa. A **larger n_0** means a **larger $\Delta \ell$** , which means **longer transmission lines**. So, there is a **tradeoff** between Δf and n_0 or, equivalently, $\Delta \ell$.

Homework

Read Chapter 9.

Homework 4 is posted and due Friday at 10 pm. I added 3 problems to the assignment compared to the initial version.

Exam 2 will be on Tuesday, June 9, from 1:30 – 3:20 pm.